German Electricity Market in 2022

M2 - Management of Economics and Business Data a.y. 2025 - 2026

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1. Introduction

The objective of this project is to investigate the relationship between renewable electricity generation and electricity prices in the German–Luxembourg electricity market during 2022. To achieve this goal, the analysis was divided into two main phases. First, SQL was used to import, verify, clean, and integrate the raw ENTSO-E datasets into a coherent analytical database. Second, R was used to performi exploratory, statistical, and econometric analyses on the resulting dataset.

The workflow follows a typical empirical research pipeline: raw data inspection, data quality assessment, dataset construction, variable engineering, descriptive analysis, and econometric modelling.

2. Database Inspection

2.1 Data sources

The first step consisted of creating tables corresponding to the available ENTSO-E datasets. Separate tables were created for electricity prices, electricity demand, renewable generation, countries, bidding zones, and the country–bidding zone correspondence mapping. The time-series tables were defined as :

```
CREATE TABLE prices_raw (
    datetime TIMESTAMPTZ,
    price NUMERIC,
    bzn TEXT
);


CREATE TABLE generation_raw (
    datetime TIMESTAMPTZ,
    generation NUMERIC,
    country TEXT,
    technology TEXT
);


CREATE TABLE demand_raw (
    datetime TIMESTAMPTZ,
    load NUMERIC,
    country TEXT
```

);

2.2 Time coverage and observational units

Once all tables had been created and populated with the imported ENTSO-E data, an initial inspection phase was carried out to gain a comprehensive understanding of the structure and characteristics of the database. This step aimed to assess the size of each dataset, identify the corresponding observational units, and evaluate the temporal coverage of the available information. In particular, the minimum and maximum timestamps were calculated for each time-series table in order to determine the exact period covered by the data and verify its suitability for the intended analysis.

Additionally, by performing this step meaningful insights for data preparation arose, since it highlighted how demand and generation were recorded using different geographical identifiers from prices table, since the firsts are referred per country while the latter per bidding zones. This information will be pivotal for the final dataset construction (particularly to specific the condition of the 'ON' clause of 'JOINS' between different time series).

3. Data Verification and Cleaning

3.1 Missing values

The next step involved a comprehensive data quality assessment. Missing values were checked across all reference and time-series tables using conditional counts. The results showed that there were not observations recorded as 'NULL' in any considered table.

However, also considered time-frequency consistency check. To do that, we analysed the spacing between consecutive timestamps to identify missing observations and verify data continuity. The demand and generation datasets exhibited regular frequencies, while the prices dataset showed a maximum gap of 3 hours instead of the expected 15 minutes, indicating the presence of missing records (that were not recorded at all or at least are totally absent as observations in this dataset).

3.2 Duplicates detection

Duplicate observations were then investigated because each dataset should have a unique observational unit. This is crucial for the subsequent construction of the empirical dataset, which will be composed by daily observations. Therefore, having multiple observations would have artificially inflated the daily counts, altering the subsequent analysis. For example, demand observations should be uniquely identified by the combination of timestamp and country, while price observations should be uniquely identified by timestamp and bidding zone. The verification revealed substantial duplication, especially in the demand and price tables, with more than 186,000 duplicated demand groups and over 37,000 duplicated price groups.

Moreover, numerical range adequacy in time series data was investigated by extracting from it the minimum value, the maximum value and by counting how many negative records existed (something unusual when dealing with prices and demand, whose minimum level should be 0). However, circa 10,000 negative prices were discovered. This represents a significant matter in which we will dive into during the empirical analysis on R, providing also insightful explanations connected to our market and period of observation.

3.3 Consistency checks

Additional consistency checks examined whether countries and bidding zones appearing in the operational datasets also existed in the reference tables. This process revealed an important inconsistency involving the German–Luxembourg electricity market. The prices table used the identifier "DE_LU", while other reference tables used either "DE-LU" or "DE".

To guarantee successful joins and maintain geographical consistency, all identifiers were harmonized under the common label "DE_LU":

```
UPDATE zones_raw SET id = 'DE_LU' WHERE id = 'DE-LU';

UPDATE generation_raw SET country = 'DE_LU' WHERE country = 'DE';

UPDATE demand_raw SET country = 'DE_LU' WHERE country = 'DE';
```

This harmonization was particularly important because Germany and Luxembourg operate within the same bidding zone and therefore constitute a single electricity market for pricing purposes. Additionally, mismatched geographical identifiers would have prevented successful joins to occur when merging observations belonging to different time series data, which would have represented a problem during our empirical analysis since the core of this project is to investigate German–Luxembourg electricity market in 2022.

3.4 Creation of Cleaned Tables

After verification, cleaned versions of the datasets were created while preserving the original raw tables. Duplicate observations were removed using the SQL window function ROW_NUMBER(), which assigns a sequential rank within each observational unit.

```
CREATE TABLE prices AS

SELECT *

FROM (

   SELECT *,

        ROW_NUMBER() OVER (

           PARTITION BY datetime, bzn

           ORDER BY datetime

        ) AS rn

   FROM prices_raw

) t

WHERE rn = 1;
```

The same procedure was applied to demand and generation tables. This approach ensured that only one observation remained for each unique timestamp–unit combination while preserving the original data for reproducibility. Moreover, a simplified version of this query was constructed also for reference tables, preserving row data while filtering them using ‘DISTINCT’ to prevent

duplicates (which were only artificially inserted in ‘cty_bzn_raw’ while harmonizing it for ‘DE_LU’ consistency).

It is worth noting that the number of deleted observations in prices exceeded the number of duplicated groups identified during verification. This occurred because some timestamp–bidding zone combinations contained more than two duplicate records, meaning that a single duplicated group could generate multiple redundant rows (as identified by the query of line 333 of the code, where there are also further comments). This aspect did not emerge in demand and generation tables, where the number of previously identified duplicated groups coincided with the number of deleted observations: for these two tables every duplicated group was composed exclusively by two observations, so just one observation per each group needed to be discarded.

4. Dataset Construction

4.1 Market Selection

The Germany–Luxembourg bidding zone was selected for several reasons. First, it is one of the largest and most liquid electricity markets in Europe, ensuring a large number of observations and highly representative market dynamics. Second, Germany has one of the highest shares of renewable electricity generation in Europe, particularly from wind and solar sources, making it an ideal case for studying the impact of renewables on electricity prices.

The analysis focuses on 2022, a particularly interesting year for European energy markets. Following the sharp increase in natural gas prices and the disruptions associated with the Russia–Ukraine conflict, electricity markets experienced unprecedented levels of price volatility and exceptionally high wholesale electricity prices. According to the Institute of Energy Economics at the University of Cologne (EWI)[1], electricity prices in Germany reached record levels during 2022, largely driven by surging gas prices and their effect on the merit-order mechanism of price formation.

4.2 Daily aggregation

Because the original datasets were recorded at a high frequency (typically every 15 minutes or every hour), the first operation consisted of aggregating all variables to a daily frequency. This aggregation was necessary because the objective of the analysis was to study medium-term relationships between market fundamentals rather than short-term intraday fluctuations.

To achieve this, three separate Common Table Expressions (CTEs) were created. The first aggregated electricity demand by summing all load observations recorded within each day:

```
daily_demand AS (
  SELECT
    DATE(datetime) AS obs_date,
    country,
```

[1] EWI (2022), *MO Tool 2022 Update*, available at: https://www.ewi.uni-koeln.de/en/news/mo-tool-2022-update/

```
    SUM(load) AS total_daily_load
  FROM demand
  WHERE country = 'DE_LU'
  GROUP BY DATE(datetime), country)
```

A second CTE performed an analogous aggregation for renewable generation, producing the total amount of renewable electricity generated each day. A third CTE aggregated electricity prices by calculating the average daily market price for the DE_LU bidding zone.

The use of separate aggregation steps offered two advantages. First, it ensured that each source dataset was transformed into a consistent daily format before integration. Second, it reduced the risk of generating duplicate observations during the subsequent joins.

4.3 Dataset Integration

Once the daily aggregates had been obtained, the three datasets were combined through a sequence of joins. The country–bidding zone correspondence table (cty_bzn) played a crucial role in this process because prices were recorded at the bidding-zone level, whereas demand and generation were recorded at the country level. The correspondence table therefore acted as a bridge between the two geographical classifications.

```
FROM cty_bzn map
JOIN daily_demand d
   ON map.cty = d.country
JOIN daily_generation g
   ON map.cty = g.country
  AND d.obs_date = g.obs_date
JOIN daily_prices p
   ON map.bzn = p.bzn
  AND d.obs_date = p.obs_date
```

The join conditions ensured that observations referred simultaneously to the same market area and the same calendar day. The resulting dataset contained one observation per day, with information on total electricity demand, total renewable generation, and average electricity prices.

4.4 Derived Variables

Several derived variables were then created to facilitate the subsequent economic analysis. The first was **residual load**, defined as total demand minus renewable generation. This variable measures the quantity of electricity that must still be supplied by conventional generators after renewable production has been taken into account.

A second derived variable was the **renewable share**, calculated as the percentage of electricity demand covered by renewable generation

```
ROUND(
   (g.total_daily_ren_gen /
    NULLIF(d.total_daily_load,0) * 100)::NUMERIC, 2
) AS ren_share_pct
```

The use of NULLIF() prevents division-by-zero errors, ensuring numerical robustness of the calculation.

Finally, a seven-day moving average of electricity prices was constructed using a SQL window function:

```
AVG(p.avg_daily_price) OVER (
   ORDER BY d.obs_date
   ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
) AS price_7d_ma
```

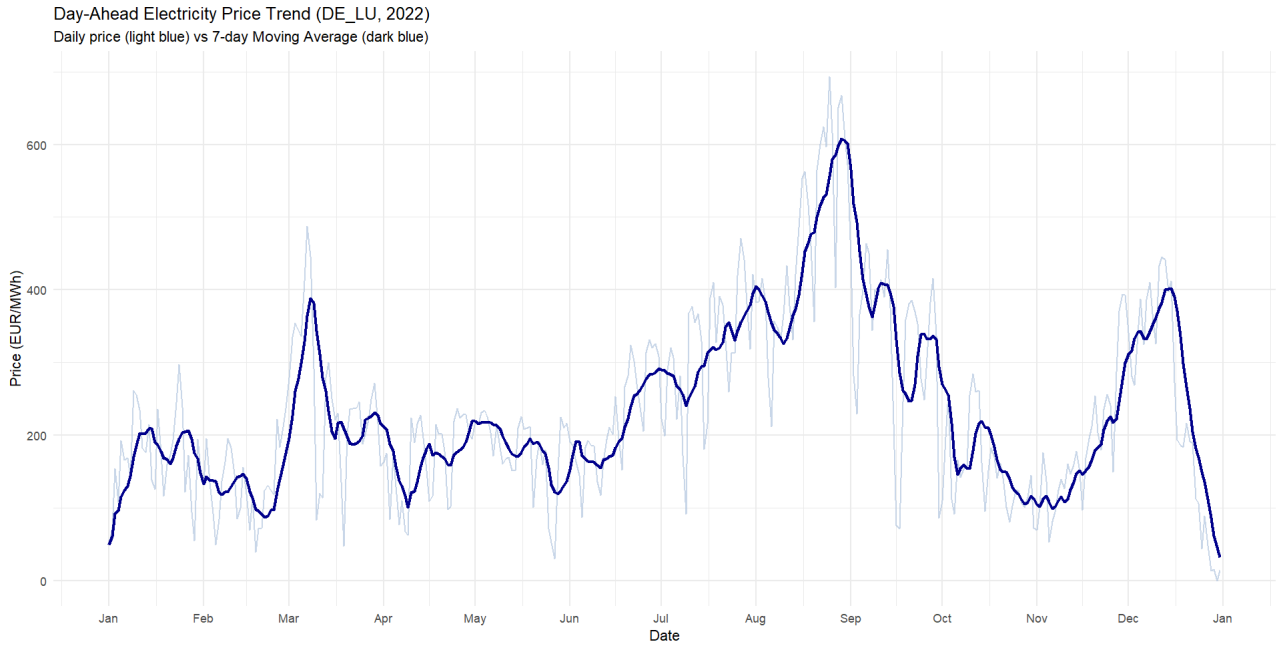
This indicator smooths short-term volatility and provides a clearer representation of underlying price trends.

As an extension, the renewable generation variable was disaggregated into its two main components, wind and solar power. This was achieved through conditional aggregation using CASE WHEN statements applied to the technology field. The resulting extended dataset contained separate measures of daily wind generation and daily solar generation, allowing the econometric analysis to investigate whether different renewable technologies exert distinct effects on electricity prices.

5. Empirical Analysis and Econometric Modeling

5.1 Wholesale Price Evolution and Macroeconomic Shocks

Figure 1 illustrates the evolution of the German-Luxembourg (DE_LU) day-ahead electricity price together with its 7-day moving average.

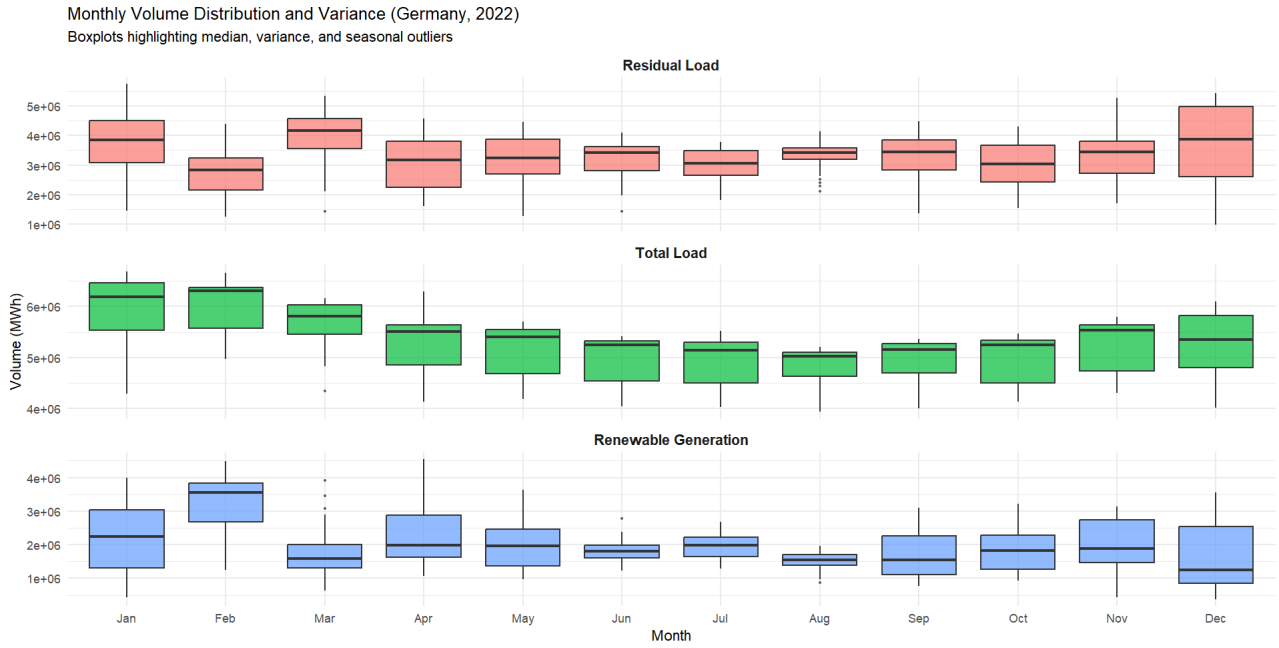


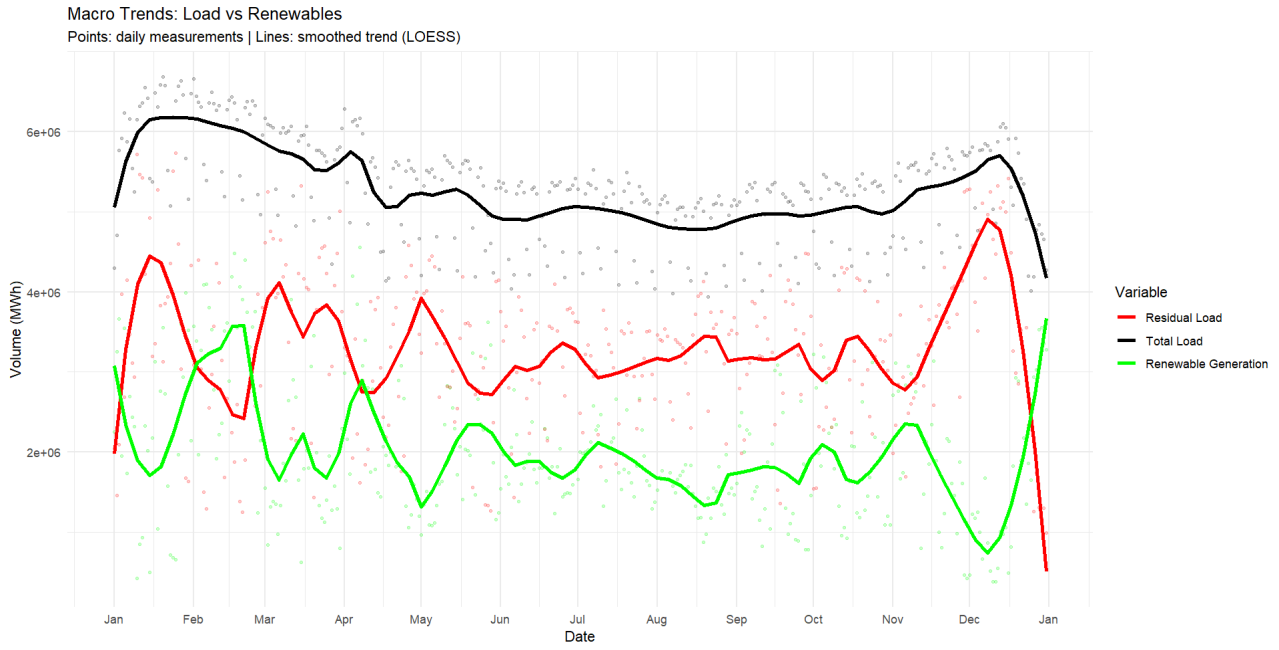
The time-series dynamics reveals the extraordinary conditions that characterized the 2022 European energy crisis, which was structurally driven by the geopolitical fallout of the Russian invasion of Ukraine. Wholesale electricity prices experienced a pronounced and persistent upward trend, frequently exceeding €200/MWh on average. The series is marked by substantial short-term volatility and several extreme price spikes. Specifically, a sharp visual surge is observable in March 2022, reflecting the initial market panic and soaring natural gas costs immediately following the outbreak of the war. This volatility escalated throughout the summer, culminating in a historic peak in late August, when Gazprom severely throttled Nord Stream 1 flows amidst a concurrent European heatwave and supply crunches, forcing daily power prices to unprecedented heights. Conversely, a dramatic price collapse is visible during the final week of December; this sharp downward trend reflects the combination of festive industrial demand destruction and a massive winter windstorm that temporarily pushed the system into an oversupply of zero-marginal-cost energy. The 7-day moving average smooths short-term volatility and removes weekly seasonal effects, such as weekend demand reductions, thereby providing a clearer view of the underlying price trend.

5.1.1. Grid Fundamentals and Asymmetric Seasonality

To better understand the physical drivers underlying electricity price dynamics, it is necessary to examine the behavior of the explanatory variables. Rather than relying on conventional time-series plots, which may suffer from overplotting and obscure distributional characteristics, we analyze the monthly distributions of the main grid volume indicators.

Figure 2 & 3 presents a set of faceted boxplots and Locally Estimated Scatterplot Smoothing, comparing the monthly distributions of total daily load (*total_daily_load*), total daily renewable generation (*total_daily_ren_gen*), and residual load (*residual_load*). This visualization allows for a direct comparison of central tendencies, dispersion, and the presence of extreme observations across the year.





The comparison reveals a pronounced structural asymmetry between the demand and supply fundamentals of the electricity market.

- **Total Grid Load (Demand).**
  Electricity demand exhibits a highly stable and predictable seasonal pattern. The monthly medians follow a characteristic U-shaped trajectory, with demand reaching its highest levels during the colder and darker months of the first and fourth quarters (Q1 and Q4), when heating and lighting requirements are elevated. Conversely, demand declines during the summer months (Q3), reaching its annual minimum. Importantly, the dispersion of observations remains relatively small and consistent throughout the year, indicating limited short-term variability in aggregate consumption.
- **Renewable Generation (Supply).**
  In contrast, renewable electricity generation displays a substantially more volatile profile.

Although solar production follows a predictable seasonal cycle, peaking during the summer months, total renewable generation is largely driven by wind power, whose output is inherently weather-dependent. As a result, renewable generation exhibits considerable intra-month variability. This pattern is particularly evident during Q1 and Q4, when stronger and more variable wind conditions generate wider interquartile ranges and a larger number of extreme observations. The distributional characteristics therefore suggest that renewable supply is governed by stochastic meteorological conditions rather than by smooth deterministic seasonal trends.

- **Residual Load (Net Demand).**
  Residual load, defined as *Residual Load = Total Load − Renewable Generation,* measures the portion of electricity demand that must be satisfied by conventional generation sources, including natural gas and coal fired power plants. Given the relative stability of total demand, fluctuations in residual load are primarily driven by variations in renewable generation. Consequently, residual load inherits much of the volatility observed in renewable output, exhibiting substantial variability throughout the year. This feature highlights the critical role of renewable intermittency in determining the demand for conventional generation and, ultimately, wholesale electricity price formation.

5.1.3. Econometric Implications of the EDA

The 2022 electricity market was characterized by high volatility due to large fluctuations in renewable generation and unusually high fossil-fuel costs during the energy crisis. Prices were driven both by domestic supply-demand factors and broader macroeconomic conditions, making simple average-based analyses insufficient. This complexity motivates an econometric approach to assess how renewable energy affected prices through the Merit Order Effect, while also accounting for external market influences.

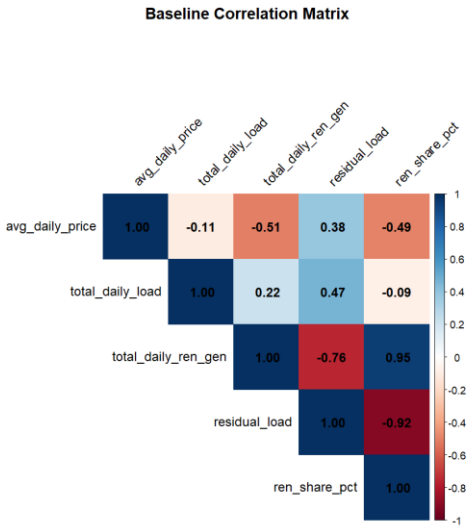
5.2. The Baseline Model: Testing the Merit Order Effect

To move beyond descriptive evidence and formally evaluate the determinants of electricity price formation, we begin by testing the central prediction of modern wholesale electricity markets: the Merit Order Effect.

In liberalized electricity markets, generation units are dispatched according to their short-run marginal costs (SRMC), with lower-cost technologies entering the market before more expensive alternatives. Because wind and solar generation have near-zero marginal operating costs, they are positioned at the bottom of the supply curve. As renewable output increases, the supply curve shifts outward, displacing higher-cost conventional generators, particularly gas- and coal-fired plants, and reducing the market-clearing price (Sensfuß, Ragwitz & Genoese, 2008). Accordingly, a higher share of renewable generation is expected to lower wholesale electricity prices, ceteris paribus. The baseline model therefore tests whether this theoretical relationship is supported by German electricity market data for 2022.

5.2.1. Preliminary Correlation Analysis

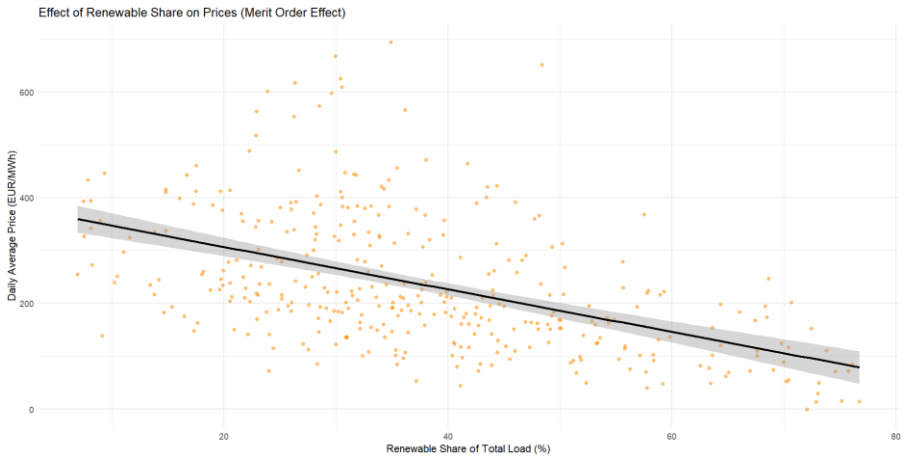
We examined the pairwise linear relationships among the key variables through a Pearson correlation matrix. Figure 4 presents the correlation matrix for daily average electricity prices, total grid load, total renewable generation, residual load, and the renewable share of total generation.



The results provide preliminary evidence consistent with economic theory.

- **Renewable Share (*ren_share_pct*).**
  It exhibits a negative correlation (-0.51) with wholesale electricity prices. This relationship suggests that periods characterized by greater penetration of renewable generation are generally associated with lower market-clearing prices, in line with the predicted Merit Order Effect.
- **Residual Load (*residual_load*).**
  Residual load displays a positive correlation (0.38) with electricity prices. Since residual load measures the portion of demand that must be met by conventional generation technologies, this result indicates that wholesale prices remain highly sensitive to the utilization of fossil-fuel-based generation, particularly during periods of elevated fuel costs.

  The negative association between renewable penetration and electricity prices is further illustrated in the scatter plot presented in Figure 5, which plots the daily average electricity price against the renewable share of generation and overlays a fitted ordinary least squares (OLS) regression line.



The negative trendline suggests a Merit Order Effect, where higher renewable generation reduces electricity prices. However, the wide dispersion shows that renewables alone do not fully explain price dynamics in 2022. Other factors such as demand, conventional generation, fuel costs, and

macroeconomic shocks also played a significant role, supporting the need for a multivariate econometric analysis.

5.2.2. Econometric Specification of the Baseline Model

To quantify the simultaneous effects of supply-side renewable generation and demand-side electricity consumption on wholesale electricity prices, we estimate the following baseline multiple linear regression model:

$$P_t = \beta_0 + \beta_1 Load_t + \beta_2 RenGen_t + \varepsilon_t$$

The specification is designed to test the central prediction of the Merit Order Effect: controlling for demand conditions, an increase in renewable electricity generation should exert downward pressure on wholesale electricity prices.

5.2.3. Baseline Empirical Results and Model Evaluation

```
Call:
lm(formula = avg_daily_price ~ total_daily_load + total_daily_ren_gen,
    data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-230.40  -75.87  -21.42   66.81  445.09

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)         3.832e+02  4.890e+01   7.837  5.2e-14 ***
total_daily_load    3.481e-07  9.398e-06   0.037     0.97
total_daily_ren_gen -7.532e-05  6.922e-06 -10.882  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 111.3 on 362 degrees of freedom
Multiple R-squared:  0.2555,    Adjusted R-squared:  0.2514
F-statistic: 62.11 on 2 and 362 DF,  p-value: < 2.2e-16
```

The baseline model was estimated using Ordinary Least Squares (OLS). The results provide preliminary empirical support for the theoretical relationship identified in the previous section.

- **Renewable Generation Coefficient.**
  The estimated coefficient associated with renewable generation is negative and highly statistically significant. This finding is consistent with the Merit Order Effect and indicates that, on average, higher levels of renewable generation are associated with lower wholesale electricity prices.
- **Goodness of Fit.**
  The model achieves a Multiple (R^2) of 25.55% and an Adjusted (R^2) of 25.14%. Thus, the two explanatory variables account for approximately one quarter of the observed variation in daily electricity prices during 2022.

Although these results provide evidence in favor of the hypothesized relationship, they also reveal important limitations of the baseline specification.

Econometric Limitations and Potential Omitted Variable Bias

While explaining 25% of daily price variation using only two physical grid variables represents a meaningful starting point, the model leaves nearly 75% of the variation in wholesale prices unexplained. This relatively modest explanatory power suggests that important determinants of electricity prices are not captured by the current specification.

Two key limitations are particularly relevant:

1. **Aggregation of Renewable Technologies.**
   Aggregating all renewables assumes a uniform impact on prices, but wind and solar differ significantly in variability and predictability, potentially masking heterogeneous effects:
   a. Solar output is concentrated during daylight hours and follows a relatively predictable seasonal pattern
   b. Wind generation is considerably more volatile and less predictable.
2. **Exclusion of External Market Conditions.**
   The model ignores external market conditions, particularly fuel price shocks during the energy crisis, which influenced electricity prices independently; this omission may cause omitted variable bias and weaken explanatory power.

Taken together, these limitations indicate that the baseline model should be viewed primarily as a benchmark specification. While it successfully identifies the expected negative relationship between renewable generation and electricity prices, it does not fully capture the complexity of price formation during an exceptionally turbulent period for European energy markets.

To address these shortcomings, the subsequent analysis extends the econometric framework by disaggregating renewable technologies.

6. Advanced Econometric Extensions

6.1. Technological Disaggregation: Wind vs. Solar Effects

To account for the differences across renewable technologies impact, total renewable generation is decomposed into its two principal components: wind generation (*wind_gen*), including both onshore and offshore production, and solar generation (*solar_gen*). This specification allows us to test whether different renewable technologies exert asymmetric effects on electricity prices.

6.1.1. Empirical Results

```
Call:
lm(formula = avg_daily_price ~ total_daily_load + wind_gen + 
    solar_gen, data = df_ext)

Residuals:
    Min      1Q  Median      3Q     Max 
-193.67  -80.48  -15.33   58.59  422.63 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)       2.256e+02  5.513e+01   4.093 5.26e-05 ***
total_daily_load  1.987e-05  9.718e-06   2.045   0.0416 *  
wind_gen         -7.436e-05  6.662e-06 -11.162  < 2e-16 ***
solar_gen         1.156e-05  1.718e-05   0.673   0.5015    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 107.1 on 361 degrees of freedom
Multiple R-squared:  0.3127,    Adjusted R-squared:  0.307 
F-statistic: 54.76 on 3 and 361 DF,  p-value: < 2.2e-16
```

The estimation of the extended model using Ordinary Least Squares (OLS) produces a notable improvement in explanatory power relative to the baseline specification.

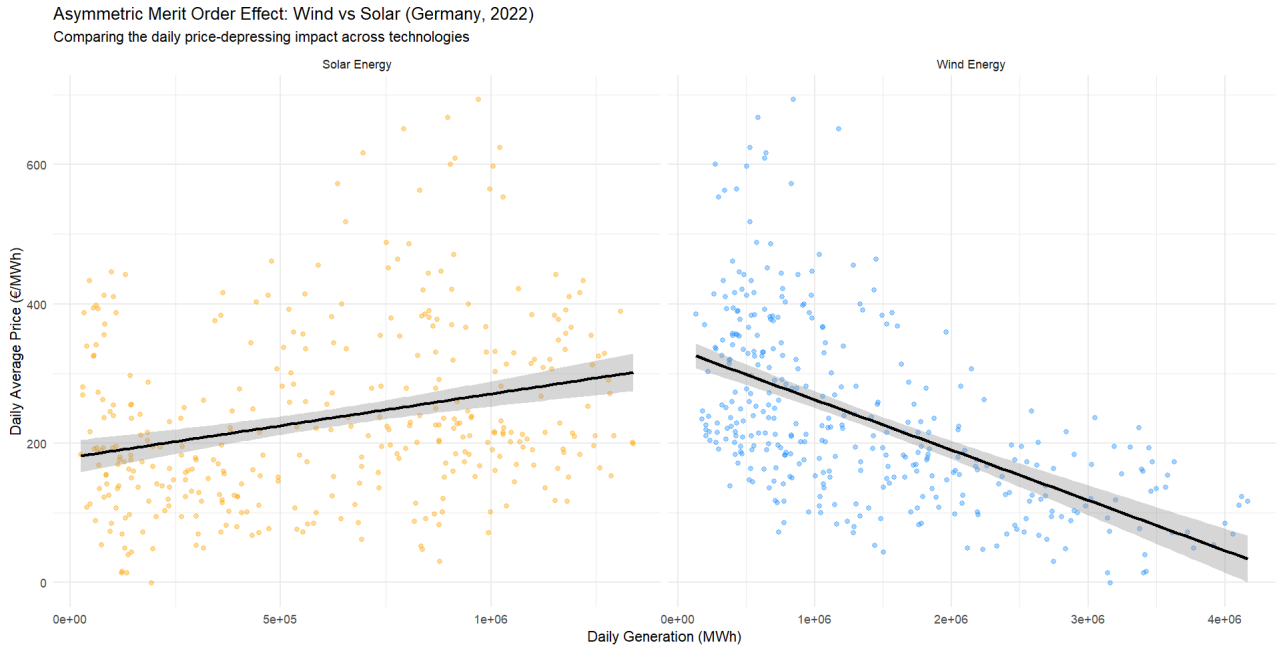
$P_t = \beta_0 + \beta_1 Load_t + \beta_2 Wind_t + \beta_3 Solar_t + \varepsilon_t$.

The results obtained from the estimation of *model_ext* reveal four key findings.

- **Improved Model Fit.**
  The Multiple (R^2) increases from 25.55% in the baseline model to 31.27%, while the Adjusted (R^2) rises to 30.70%. This improvement indicates that distinguishing between renewable technologies captures additional variation in electricity prices and provides a more accurate representation of market dynamics. The increase in explanatory power supports the hypothesis that renewable technologies affect electricity prices in different ways.
- **The Dominant Role of Wind Generation).**
  Wind generation exhibits a negative and highly statistically significant coefficient. This result provides strong evidence that wind generation contributes substantially to reducing wholesale electricity prices. In economic terms, an increase of 10,000 MWh (10 GWh) in daily wind production is associated with a reduction of approximately €0.74/MWh in the daily average electricity price, holding all other factors constant. The magnitude and significance of this effect suggest that wind generation was the primary driver of the observed Merit Order Effect during 2022.
- **The Limited Impact of Solar Generation.**
  In contrast, the coefficient associated with solar generation is positive but statistically insignificant Consequently, there is insufficient evidence to conclude that variations in daily solar generation exert a systematic effect on average day-ahead electricity prices. While solar power undoubtedly contributes to overall electricity supply, its impact appears less pronounced when measured at the daily level.
- **The Re-emergence of Demand Effects.**
  After controlling separately for wind and solar generation, total electricity demand becomes statistically significant at the 5% level. This finding is consistent with standard economic theory: holding supply conditions constant, higher demand increases the need for additional generation capacity and places upward pressure on wholesale prices. The result suggests that

the aggregate renewable measure used in the baseline model may have partially obscured the underlying influence of demand on price formation.

Figure 6 presents scatter plots of daily electricity prices against wind and solar generation, together with the corresponding fitted regression lines, highlighting the contrasting relationships between the two technologies and wholesale market prices.

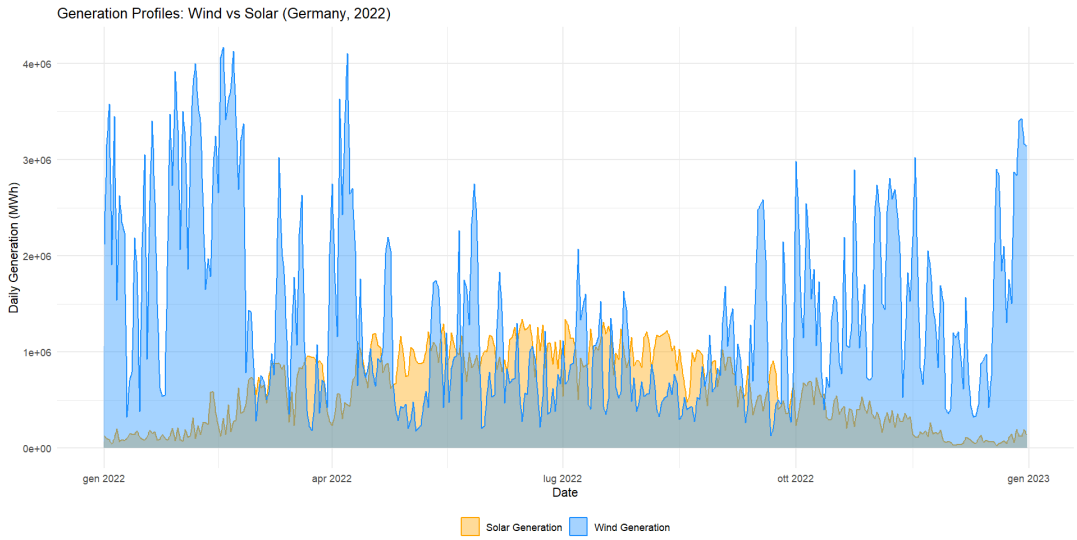


Overall, the extended specification provides a more detailed understanding of the mechanisms driving electricity prices. The results demonstrate that the aggregate renewable effect identified in the baseline model is largely attributable to wind generation, while solar generation does not exhibit a statistically significant impact at the daily level. Furthermore, the improved model fit highlights the importance of accounting for technological heterogeneity when analyzing the economic effects of renewable energy integration.

6.1.2. Economic Interpretation of the Asymmetry

The statistical findings found about different impact of wind and solar generation can be explained by the distinct production profiles of these technologies and their interaction with electricity demand patterns.

Figure 7 shows the volume of generated power (MWh) between solar and wind during the whole year of 2022.



Solar Generation and Demand Alignment.
Solar photovoltaic generation is concentrated during daylight hours, typically reaching its maximum output around midday. This production profile closely coincides with periods of elevated electricity demand driven by commercial and industrial activity. As a result, solar generation tends to enter the market when consumption levels and willingness to pay are relatively high. The additional renewable supply can be absorbed more easily by the system, reducing peak prices without fundamentally altering the overall market equilibrium. Consequently, solar generation primarily moderates periods of high prices rather than exerting a substantial influence on average daily prices.

Wind Generation and Off-Peak Supply Shocks.
Wind generation, by contrast, is largely independent of human consumption patterns and is determined by meteorological conditions. Electricity produced by wind farms can enter the grid at any hour of the day, including periods of low demand such as nighttime hours or weekends. During episodes of strong wind generation, particularly when demand is subdued, large quantities of low-marginal-cost electricity may displace conventional thermal generation from the market.

Given the non-storability of electricity at scale and the requirement for continuous real-time balancing, large wind injections can significantly reduce wholesale electricity prices by displacing conventional generators. As the dependent variable is the daily average price, these low-price periods exert a strong influence on the daily mean, reinforcing the negative relationship between wind generation and wholesale prices.

6.2. Nonlinear Dynamics and the Price Cannibalization Effect

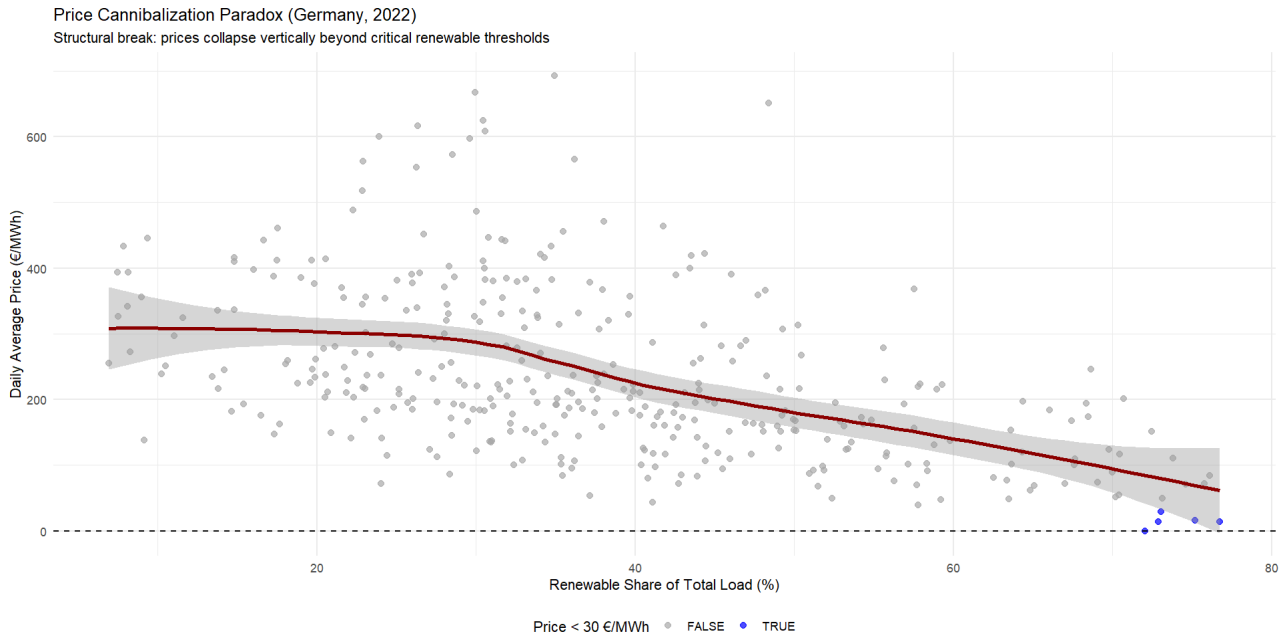
Although the OLS model confirms that renewable generation reduces electricity prices, it assumes that this effect remains constant across all market conditions. Electricity markets are highly nonlinear, and the price impact of additional renewable generation may intensify at high levels of penetration.

This phenomenon, known as the **Price Cannibalization Effect**, occurs when large volumes of renewable generation depress wholesale prices precisely during periods of peak production, thereby reducing the market value of renewable output. The following analysis investigates whether such nonlinear dynamics are evident in the German electricity market during 2022.

6.2.1. The Economics of Price Cannibalization

The Price Cannibalization Effect arises because renewable generators are exposed to similar weather conditions and therefore tend to produce electricity simultaneously. During periods of favorable weather, large volumes of low-marginal-cost electricity enter the market, shifting the supply curve outward and placing downward pressure on wholesale prices.

As renewable penetration increases, this mechanism can reduce the average price received by renewable producers, creating a trade-off between higher renewable output and lower market value. Figure 8 examines this relationship by plotting daily average electricity prices against the renewable share of generation and fitting a non-parametric LOESS curve to capture potential nonlinearities.



The LOESS curve indicates a nonlinear relationship between renewable penetration and electricity prices. For shares below roughly 30–35%, prices remain relatively stable, while beyond this threshold the price-reducing effect becomes much stronger.

This suggests a structural change in price formation: once renewable supply is sufficiently high, additional generation has a disproportionately large downward impact on prices, consistent with the Price Cannibalization Effect.

6.2.2. Tail-Event Case Study: The December 2022 Price Collapse

To further investigate the mechanisms underlying this nonlinear behavior, we examine a set of extreme observations corresponding to days on which the daily average electricity price fell below €30/MWh.

	obs_date	avg_daily_price	ren_share_pct	wind_gen
	<date>	<dbl>	<dbl>	<dbl>
1	2022-12-30	-0.764	72.1	3162799.
2	2022-12-28	13.6	72.9	3405338.
3	2022-12-31	14.1	76.8	3141996.
4	2022-12-29	15.5	75.2	3428006.
5	2022-05-27	29.5	73.1	2746117.

The extracted observations reveal a clear clustering of low-price events at the end of December 2022, particularly during the period from 28 to 31 December. These days represent an exceptional combination of supply and demand conditions that created an ideal environment for extreme price reductions.

Two factors were particularly important:

1. **Reduced Electricity Demand.**
   The period between Christmas and New Year is characterized by a substantial decline in industrial and commercial activity, making aggregate electricity demand falls significantly below normal seasonal levels.

2. **Exceptional Wind Generation.**
   At the same time, Germany experienced unusually strong wind conditions. Daily wind generation consistently exceeded 3 million MWh and reached some of the highest values observed during the year, covering more than 70% of total electricity supply.

The most striking observation occurred on 30 December 2022, when the German electricity market recorded a negative daily average price of –€0.764/MWh while renewable generation represented 72.1% of total supply. Given that wholesale electricity prices had frequently exceeded €200/MWh during the energy crisis, the occurrence of a negative daily average price constitutes an extraordinary market outcome.

The Economic Logic of Negative Prices
Negative electricity prices arise from operational constraints in conventional power plants, which may continue operating during periods of excess supply to avoid costly shutdowns and restarts. When high renewable output coincides with very low demand, these dynamics can push market prices below zero. The December 2022 episode illustrates this mechanism and provides a clear example of the Price Cannibalization Effect.

6.2.3. Categorical Verification: Market Regime Analysis

To further validate the nonlinear relationship identified in the previous section, the dataset was partitioned into three distinct market regimes based on the level of renewable penetration. This approach enables a comparison of average market outcomes under different renewable-generation conditions and provides an intuitive framework for assessing the magnitude of the Price Cannibalization Effect.

Macroeconomic Market Regimes Summary (2022)

	Regime	Days	Avg_Price_Euro	Avg_Wind_MWh
	<chr>	<int>	<dbl>	<dbl>
1	1. Low Penetration (<30%)	119	296.	626537
2	2. Medium Penetration (30-60%)	211	224.	1511020
3	3. High Penetration (>60%)	35	99.4	3126989

The regime-based analysis provides strong empirical evidence of the relationship between renewable penetration and wholesale electricity prices: when low penetration regime of renewable share, fossil-fuel plants remain the marginal producers, leading to very high prices due to elevated fuel costs during the 2022 crisis (around €296/MWh). When renewable penetration is high (renewable share > 60%), increased wind generation displaces conventional production, sharply reducing prices to about €99.40/MWh (a drop of roughly 66%).

These results highlight the dual effect of renewable expansion: while it significantly benefits consumers by lowering average prices, it also reduces the revenues of renewable producers as supply increases, capturing the core trade-off underlying the Price Cannibalization Effect.

7. Model Refinement and Geopolitical Contextualization

7.1. Incorporating Monthly Fixed Effects

The models estimated so far focus on internal grid fundamentals, namely electricity demand, wind generation, and solar generation. While these variables capture key supply-and-demand dynamics, the extended model explains only about 31% of electricity price variation, suggesting the presence of important omitted factors.

In 2022, wholesale electricity prices were heavily influenced by external forces, including fuel price volatility, geopolitical uncertainty, and policy interventions associated with the European energy crisis. As gas-fired plants often set marginal price, changes in natural gas costs affected electricity prices independently of domestic grid conditions. Rather than modelling these factors directly, monthly fixed effects are introduced to control for unobserved time-varying influences in a parsimonious way.

7.1.1. Econometric Specification with Time Fixed Effects

The inclusion of monthly dummy variables allows the intercept of the regression model to vary across months, capturing systematic differences in the average price level over time. By accounting for these month-specific effects, the model isolates the contribution of daily demand and renewable generation more accurately, reducing the risk of omitted variable bias and improving the precision of the estimated coefficients.

7.1.2. Empirical Results and Improved Explanatory Power

```
Call:
lm(formula = avg_daily_price ~ total_daily_load + wind_gen + 
    solar_gen + month_factor, data = df_ext)

Residuals:
    Min      1Q  Median      3Q     Max 
-268.64  -47.13   -5.62   41.77  245.94 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)       8.427e+01  4.789e+01   1.760  0.07936 .  
total_daily_load  4.748e-05  7.968e-06   5.959 6.20e-09 ***
wind_gen         -5.700e-05  5.268e-06 -10.820  < 2e-16 ***
solar_gen        -3.599e-05  2.364e-05  -1.522  0.12885    
month_factor.L    9.406e+01  1.697e+01   5.541 5.91e-08 ***
month_factor.Q   -1.363e+02  2.978e+01  -4.578 6.53e-06 ***
month_factor.C   -8.967e+01  1.404e+01  -6.386 5.43e-10 ***
month_factor^4    7.394e+01  1.455e+01   5.082 6.10e-07 ***
month_factor^5    1.705e+02  1.383e+01  12.324  < 2e-16 ***
month_factor^6    2.416e+01  1.427e+01   1.693  0.09126 .  
month_factor^7   -1.187e+02  1.410e+01  -8.416 1.01e-15 ***
month_factor^8   -5.781e+01  1.396e+01  -4.141 4.33e-05 ***
month_factor^9   -1.061e+01  1.395e+01  -0.761  0.44738    
month_factor^10   3.821e+01  1.391e+01   2.748  0.00631 ** 
month_factor^11   1.749e+01  1.368e+01   1.278  0.20198    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 75.52 on 350 degrees of freedom
Multiple R-squared:  0.6686,    Adjusted R-squared:  0.6554 
F-statistic: 50.44 on 14 and 350 DF,  p-value: < 2.2e-16
```

- **Improved Goodness-of-Fit.**
  The model achieves a Multiple (R^2) of 66.86% and an Adjusted (R^2) of 65.54%, representing a considerable increase relative to the extended model, which explained 31.26% of the variation in daily electricity prices. The inclusion of monthly fixed effects therefore increases the explained variance by more than 35 percentage points. This result suggests that a substantial share of electricity price fluctuations during 2022 was driven by time-varying factors operating at the monthly level, beyond the influence of daily demand and renewable generation patterns.
- **Enhanced Predictive Accuracy.**
  The Residual Standard Error decreases from 107.1 to 75.53, indicating a significant reduction in the average magnitude of prediction errors. This improvement demonstrates that controlling for month-specific effects substantially enhances the model's ability to capture observed price dynamics.
- **Demand Effects (total_daily_load).**
  After accounting for monthly fixed effects, the coefficient associated with total electricity demand increases and becomes highly statistically significant. This finding provides strong evidence that demand conditions exert an important influence on wholesale electricity prices once broader temporal effects are controlled for.
- **Wind Generation Effects**
  The coefficient on wind generation remains negative and highly significant as in previous models, confirm the central role of wind generation in reducing wholesale electricity prices through the Merit Order Effect.

Solar Generation Effects ($\beta 3 Solar_t$)
Consistent with the results of the previous model, solar generation remains statistically

insignificant, suggesting that the limited impact of solar generation on average daily electricity prices cannot be attributed solely to omitted monthly factors. Rather, it reflects the underlying characteristics of solar production and its interaction with demand patterns.

Overall, the inclusion of monthly fixed effects significantly improves the explanatory power of the model while reinforcing the key findings obtained from earlier specifications. In particular, demand and wind generation emerge as robust determinants of electricity prices, whereas solar generation continues to exhibit no statistically significant effect at daily frequency.

7.1.3. Interpretation of Monthly Fixed Effects

In the R output, monthly fixed effects are represented through orthogonal polynomial contrasts, which efficiently capture systematic temporal variation while limiting multicollinearity. The significance of the linear, quadratic, and cubic terms indicates that electricity prices followed structured patterns throughout 2022 rather than random fluctuations. By accounting for these month-specific effects, the model more effectively isolates the impact of demand and renewable generation, leading to more precise coefficient estimates.

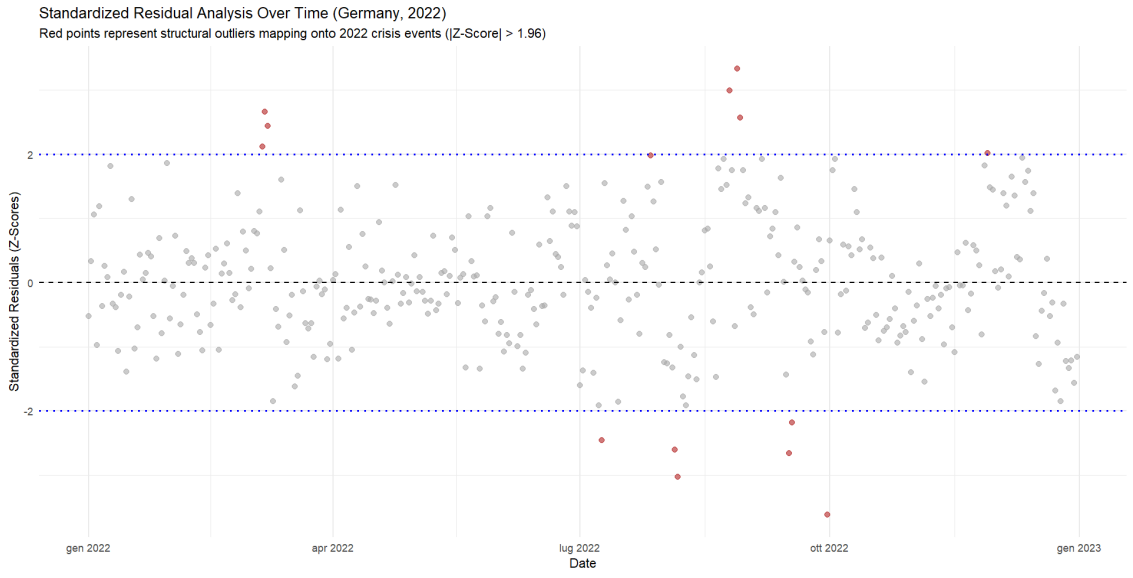
7.2. Standardized Residuals and the 2022 Crisis Timeline

Although the fixed-effects model explains a substantial share of price variation, residual analysis remains essential for identifying periods in which the model performs poorly. To ensure comparability across time, standardized residuals are used instead of raw residuals, providing a scale-independent measure of forecasting errors and helping identify unusual events, structural breaks, or model misspecification.

7.2.1. Methodological Justification for Standardized Residuals

A raw residual measures the difference between the observed and predicted electricity price. To obtain a scale-independent metric, residuals are standardized by dividing them by the residual standard error, producing values comparable to z-scores. Under standard assumptions, observations exceeding ±1.96 are considered statistically unusual and may indicate events not captured by the model.

Figure 9 presents the standardized residuals over time together with the critical thresholds at ±1.96.



7.2.2. Historical Interpretation of Statistical Outliers

The distribution of standardized residuals reveals that large forecasting errors are not randomly dispersed throughout the sample period. Instead, they appear in distinct temporal clusters that coincide with major geopolitical and market events affecting the European energy sector during 2022.

1. **March 2022: The Initial Geopolitical Shock**
   Large positive residuals appear in late February / early March 2022, immediately after the Russian invasion of Ukraine, reflecting the sharp disruption in European energy markets. During this period, extreme volatility in natural gas prices, driven by uncertainty over supply, strongly affected electricity prices, as gas-fired plants often set the marginal price. The model produces positive residuals because domestic demand and renewable generation cannot explain such large price increases; instead, prices incorporated a geopolitical risk premium originating outside the domestic system.
2. **August - September 2022: The Peak of the Energy Crisis**
   The largest positive residuals occur in August–early September 2022, during the peak of the European energy crisis, when multiple adverse shocks significantly drove electricity prices above model predictions:
   - Natural Gas Supply Constraints. Reduced flows through Nord Stream 1 increased concerns about energy security and pushed gas prices to record levels, raising electricity prices.
   - Low Rhine River Water Levels. Drought conditions disrupted fuel transport, limiting coal deliveries and increasing supply concerns.

   These combined shocks led to electricity prices far above levels explained by domestic demand and renewables, with standardized residuals exceeding critical thresholds. This indicates that abnormal external factors, rather than typical grid fundamentals, were driving price dynamics.
3. **December 2022: Extreme Negative Price Deviations**
   A cluster of large negative residuals appears in the final week of December, corresponding to the period of price cannibalization. During the Christmas holidays, industrial demand dropped significantly while exceptionally strong wind generation produced abundant low-cost electricity. This combination pushed wholesale prices far below the levels predicted by the model. Since the fixed effects already capture typical December patterns, these extreme conditions generated strongly negative standardized residuals, with several observations falling below the −1.96 threshold.

7.3. Joint Supply - Demand Interaction Analysis

We extend the analysis by examining the interaction between renewable supply and electricity demand. While previous models considered these factors separately, economic theory implies that prices depend on their interaction. The impact of renewable shocks varies with demand conditions, making this interaction essential for understanding when renewables have the strongest price effects.

7.3.1. Methodological Approach & Emprical Results

High wind days are defined as observations above the 80th percentile of annual wind generation. These are split into two regimes based on median electricity demand (~1.35 million MWh):

- High Wind–High Demand: wind above 80th percentile, demand above median;
- High Wind–Low Demand: wind above 80th percentile, demand below median.

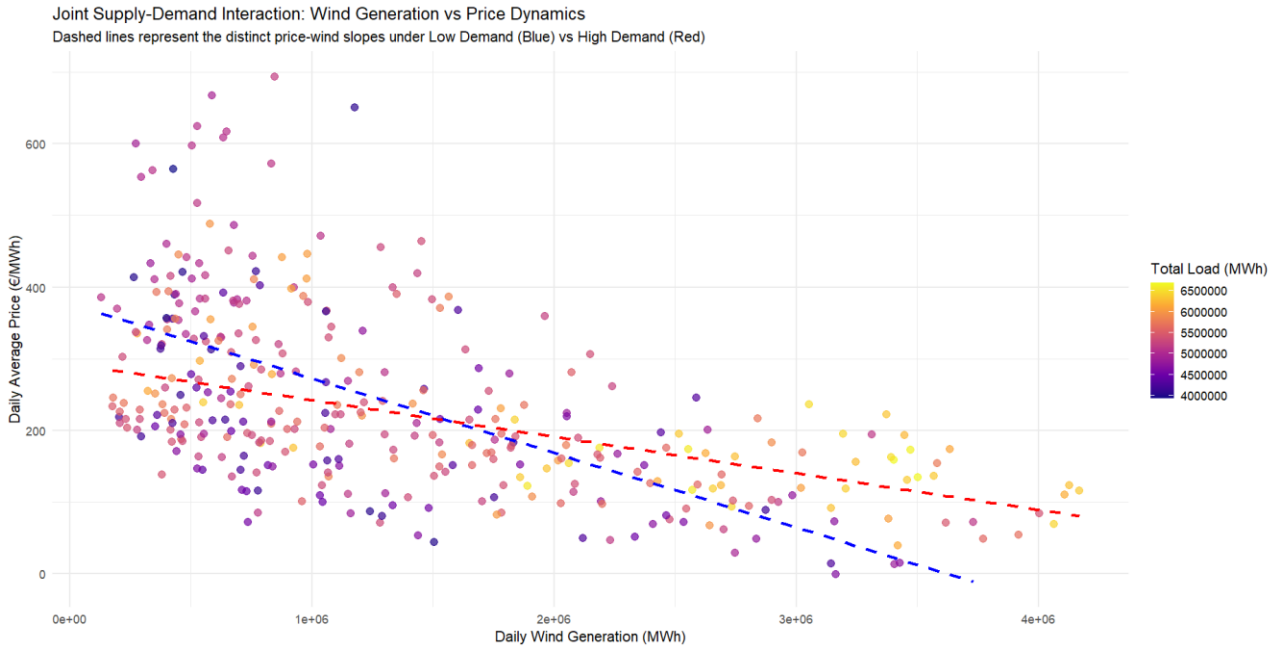
Comparing prices across these regimes reveals how demand moderates renewable price effects.

- **High Wind–High Demand State.** When high wind coincides with strong demand, the system absorbs renewable supply without extreme price drops. Conventional generation still operates, keeping prices positive and relatively stable.

- **High Wind–Low Demand State**. When high wind occurs during weak demand, renewable supply dominates, sharply reducing prices and sometimes causing negative prices. Overall, extreme price reductions arise when high renewable output coincides with low demand.

7.3.2. Visual Interpretation of the Interaction Effect

Figure 10 illustrates the relationship between wind generation and electricity prices, distinguishing between high- and low-demand conditions with separate fitted lines.



- High-Demand Regime. The relationship has a moderate negative slope: higher wind reduces prices, but strong demand limits the effect by maintaining the role of conventional generation.

- Low-Demand Regime. The slope is much steeper and lower: under weak demand, additional wind strongly depresses prices, increasing the likelihood of very low or negative prices.

Overall, the difference across regimes shows that the price effect of wind is not constant but depends on demand conditions.

Conclusions

The results show that extreme price dynamics in 2022 arise from the interaction of renewable supply and demand. Large renewable shocks alone do not cause price collapses; these occur mainly when combined with weak demand. Thus, the Merit Order Effect is state-dependent. From a policy perspective, this highlights the importance of demand flexibility, storage, and system balancing to manage high renewable penetration and limit price cannibalization.

From a policy perspective, these results underscore the importance of demand flexibility, energy storage, and system-balancing mechanisms in highly decarbonized electricity systems. As renewable penetration continues to increase, the ability of the power system to absorb periods of abundant renewable generation will become increasingly important for maintaining market stability and mitigating the adverse effects of price cannibalization.

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